DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE; THE FACILITY OWNER'S QUALIFIED PROCEDURES AND APPLICABLE CODES GOVERN.

WO-100-01 — Internal inspection of tank T-101

Area	A100 Tank farm	**Task**	TI Tank internal inspection
Priority	2 (integrity)	**Issued by**	Facility owner (demonstration)
Primary asset	T-101 — T-101 tank 12' dia × 22' high — E 25'-0" N 26'-6" EL 100'-6"		
Equipment	T-101	**Lines**	6"-P150-P-1001, 4"-P150-P-6038, 1"-P150-V-6030, 1"-P150-D-6031, 1"-P150-N-3015, 2"-P300-LS-3007, 2"-P150-SC-3008, 3/4"-P150-I-6032, 3/4"-P150-I-6033
Isometrics	ISO-P-1001, ISO-P-6038, ISO-V-6030, ISO-D-6031, ISO-N-3015, ISO-LS-3007, ISO-SC-3008, ISO-I-6032, ISO-I-6033 (owner's drawings, attached)		
Permits	General work permit, Confined-space entry permit		

Reason

Scheduled internal inspection of the shell, floor and roof.

Scope of work

1. Isolate T-101 per the isolation list; drain, vent, purge and gas-test for entry.
2. Open for entry; clean as needed for inspection.
3. Internal visual examination; ultrasonic thickness at the 15 TMLs; floor scan.
4. Close with new gaskets on every opened joint; remove blinds; restore the line-up.

Affected items: T-101

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-TEST-2, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-5, SW-WO-6, SW-WO-7

Owner's isometric drawings (SW-IMP-6)

Drawing	Line drawn	Work line	Sheets	Match to the plant model
ISO-P-1001	6"-P150-P-1001	6"-P150-P-1001	5	1 error(s), 4 warning(s) — see below
ISO-P-6038	4"-P150-P-6038	4"-P150-P-6038	1	agrees with the model
ISO-V-6030	1"-P150-V-6030	1"-P150-V-6030	1	agrees with the model
ISO-D-6031	1"-P150-D-6031	1"-P150-D-6031	1	agrees with the model
ISO-N-3015	1"-P150-N-3015	1"-P150-N-3015	1	0 error(s), 1 warning(s) — see below
ISO-LS-3007	2"-P300-LS-3007	2"-P300-LS-3007	1	agrees with the model
ISO-SC-3008	2"-P150-SC-3008	2"-P150-SC-3008	1	0 error(s), 2 warning(s) — see below
ISO-I-6032	3/4"-P150-I-6032	3/4"-P150-I-6032	1	agrees with the model
ISO-I-6033	3/4"-P150-I-6033	3/4"-P150-I-6033	1	agrees with the model

ISO-P-1001 (PDF): ../isometrics-imported/ISO-P-1001.pdf · ISO-P-6038 (PDF): ../isometrics-imported/ISO-P-6038.pdf · ISO-V-6030 (PDF): ../isometrics-imported/ISO-V-6030.pdf · ISO-D-6031 (PDF): ../isometrics-imported/ISO-D-6031.pdf · ISO-N-3015 (PDF): ../isometrics-imported/ISO-N-3015.pdf · ISO-LS-3007 (PDF): ../isometrics-imported/ISO-LS-3007.pdf · ISO-SC-3008 (PDF): ../isometrics-imported/ISO-SC-3008.pdf · ISO-I-6032 (PDF): ../isometrics-imported/ISO-I-6032.pdf · ISO-I-6033 (PDF): ../isometrics-imported/ISO-I-6033.pdf — the sheets that hold the work follow the ITP.

Drawing	Sheet	Level	Finding
ISO-P-1001	5	error	sheet 5: CONT'D ON ISO-P-1002 prints E 154'-5" N -12'-6" EL 102'-0" for E 154'-5" N -11'-6" EL 102'-0": the minus sign applies to the feet only; read as written the point is 12.00 in away and 12.00 in off where 6"-P150-P-1001 meets 6"-P150-P-1002 (E 154'-5" N -11'-6" EL 102'-0")
ISO-P-1001	5	warning	W21 (6"-P150-P-1001/W21, sheet 5) is labelled again on sheet 4; it is 58.0 in from the nearest break
ISO-P-1001	4	warning	branch weld 3/4"-P300-I-7030/W1 (E 151'-6" N 0" EL 117'-5", sheet 4) has no label
ISO-P-1001	4	warning	branch weld 3/4"-P300-I-7001/W1 (E 152'-6" N 0" EL 117'-5", sheet 4) has no label
ISO-P-1001	5	warning	TO label 'P-101A NOZ. S' is not on the drawing
ISO-N-3015	1	warning	weld 1"-P150-N-3015/W1 (E 28'-0" N 3'-6" EL 122'-8", sheet 1) has no label
ISO-SC-3008	1	warning	weld 2"-P150-SC-3008/W11 (E 23'-6" N -1'-8 11/16" EL 125'-0", sheet 1) has no label
ISO-SC-3008	1	warning	weld 2"-P150-SC-3008/W1 (E 23'-6" N -2'-0" EL 122'-8 1/2", sheet 1) has no label

Assets in the plant model (SW-WO-7)

Tag	Kind	Description	Line	Location
T-101	equipment	T-101 tank 12' dia × 22' high		E 25'-0" N 26'-6" EL 100'-6"
6"-P150-P-1001/A2	item	VALVE, GATE, WEDGE, OS&Y 6" P150	6"-P150-P-1001	E 32'-2-15/16" N 26'-6" EL 101'-6"
T-101-N1	nozzle	Nozzle T-101-N1 (outlet, 6" P150) on T-101		E 31'-7-13/16" N 26'-6" EL 101'-6"
4"-P150-P-6038/2	item	VALVE, GATE, WEDGE, OS&Y 4" P150	4"-P150-P-6038	E 25'-0" N 22'-6" EL 123'-10-13/16"

Tag	Kind	Description	Line	Location
T-101-N2	nozzle	Nozzle T-101-N2 (inlet, 4" P150) on T-101		E 25'-0" N 22'-6" EL 123'-3-1/16"
3"-P150-RV-4002/2	item	VALVE, PRESSURE SAFETY VALVE 3" P150 INLET × 4" P150 OUTLET	3"-P150-RV-4002	E 22'-0" N 26'-6" EL 124'-0"
1"-P150-V-6030/2	item	VALVE, BALL, FLOATING 1" P150	1"-P150-V-6030	E 25'-0" N 26'-6" EL 123'-10-1/2"
T-101-N4	nozzle	Nozzle T-101-N4 (vent, 1" P150) on T-101		E 25'-0" N 26'-6" EL 123'-3-11/16"
1"-P150-D-6031/2	item	VALVE, BALL, FLOATING 1" P150	1"-P150-D-6031	E 18'-0" N 26'-6" EL 101'-0"
T-101-N5	nozzle	Nozzle T-101-N5 (drain, 1" P150) on T-101		E 18'-6-7/8" N 26'-6" EL 101'-0"
1"-P150-N-3015/B2	item	VALVE, BALL, FLOATING 1" P150	1"-P150-N-3015	E 28'-0" N 26'-6" EL 123'-8-1/4"
T-101-N6	nozzle	Nozzle T-101-N6 (nitrogen, 1" P150) on T-101		E 28'-0" N 26'-6" EL 123'-1-7/16"
2"-P300-LS-3007/B2	item	VALVE, GATE, WEDGE, OS&Y 2" P300	2"-P300-LS-3007	E 18'-5/8" N 26'-6" EL 102'-6"
T-101-N7	nozzle	Nozzle T-101-N7 (steam in, 2" P300) on T-101		E 18'-6-1/8" N 26'-6" EL 102'-6"
2"-P150-SC-3008/A2	item	VALVE, GATE, WEDGE, OS&Y 2" P150	2"-P150-SC-3008	E 18'-5/8" N 26'-6" EL 104'-6"
T-101-N8	nozzle	Nozzle T-101-N8 (condensate out, 2" P150) on T-101		E 18'-6-1/8" N 26'-6" EL 104'-6"
3/4"-P150-I-6032/2	item	VALVE, BALL, FLOATING 3/4" P150	3/4"-P150-I-6032	E 18'-1/4" N 26'-6" EL 107'-0"
T-101-N9	nozzle	Nozzle T-101-N9 (instrument, 0.75" P150) on T-101		E 18'-7" N 26'-6" EL 107'-0"
3/4"-P150-I-6033/2	item	VALVE, BALL, FLOATING 3/4" P150	3/4"-P150-I-6033	E 18'-1/4" N 26'-6" EL 117'-0"
T-101-N10	nozzle	Nozzle T-101-N10 (instrument, 0.75" P150) on T-101		E 18'-7" N 26'-6" EL 117'-0"

Isolation list (SW-WO-3)

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
1	valve	6"-P150-P-1001/A2	6"-P150-P-1001	Close and lock VALVE, GATE, WEDGE, OS&Y 6" P150 (6"-P150-P-1001/A2) on T-101-N1	E 32'-2-15/16" N 26'-6" EL 101'-6"		
2	blind	T-101-N1	6"-P150-P-1001	Fit a blind between 6"-P150-P-1001/A2 and the nozzle flange T-101-N1	E 31'-7-13/16" N 26'-6" EL 101'-6"		
3	valve	4"-P150-P-6038/2	4"-P150-P-6038	Close and lock VALVE, GATE, WEDGE, OS&Y 4" P150 (4"-P150-P-6038/2) on T-101-N2	E 25'-0" N 22'-6" EL 123'-10-13/16"		
4	blind	T-101-N2	4"-P150-P-6038	Fit a blind between 4"-P150-P-6038/2 and the nozzle flange T-101-N2	E 25'-0" N 22'-6" EL 123'-3-1/16"		
5	removal	3"-P150-RV-4002/2	3"-P150-RV-4002	Remove the relief valve on T-101-N3 and blind the nozzle			
6	valve	1"-P150-V-6030/2	1"-P150-V-6030	Close and lock VALVE, BALL, FLOATING 1" P150 (1"-P150-V-6030/2) on T-101-N4	E 25'-0" N 26'-6" EL 123'-10-1/2"		
7	blind	T-101-N4	1"-P150-V-6030	Fit a blind between 1"-P150-V-6030/2 and the nozzle flange T-101-N4	E 25'-0" N 26'-6" EL 123'-3-11/16"		
8	valve	1"-P150-D-6031/2	1"-P150-D-6031	Close and lock VALVE, BALL, FLOATING 1" P150 (1"-P150-D-6031/2) on T-101-N5	E 18'-0" N 26'-6" EL 101'-0"		
9	blind	T-101-N5	1"-P150-D-6031	Fit a blind between 1"-P150-D-6031/2 and the nozzle flange T-101-N5	E 18'-6-7/8" N 26'-6" EL 101'-0"		
10	valve	1"-P150-N-3015/B2	1"-P150-N-3015	Close and lock VALVE, BALL, FLOATING 1" P150 (1"-P150-N-3015/B2) on T-101-N6	E 28'-0" N 26'-6" EL 123'-8-1/4"		
11	blind	T-101-N6	1"-P150-N-3015	Fit a blind between 1"-P150-N-3015/B2 and the nozzle flange T-101-N6	E 28'-0" N 26'-6" EL 123'-1-7/16"		
12	valve	2"-P300-LS-3007/B2	2"-P300-LS-3007	Close and lock VALVE, GATE, WEDGE, OS&Y 2" P300 (2"-P300-LS-3007/B2) on T-101-N7	E 18'-5/8" N 26'-6" EL 102'-6"		
13	blind	T-101-N7	2"-P300-LS-3007	Fit a blind between 2"-P300-LS-3007/B2 and the nozzle flange T-101-N7	E 18'-6-1/8" N 26'-6" EL 102'-6"		
14	valve	2"-P150-SC-3008/A2	2"-P150-SC-3008	Close and lock VALVE, GATE, WEDGE, OS&Y 2" P150 (2"-P150-SC-3008/A2) on T-101-N8	E 18'-5/8" N 26'-6" EL 104'-6"		
15	blind	T-101-N8	2"-P150-SC-3008	Fit a blind between 2"-P150-SC-3008/A2 and the nozzle flange T-101-N8	E 18'-6-1/8" N 26'-6" EL 104'-6"		
16	valve	3/4"-P150-I-6032/2	3/4"-P150-I-6032	Close and lock VALVE, BALL, FLOATING 3/4" P150 (3/4"-P150-I-6032/2) on T-101-N9	E 18'-1/4" N 26'-6" EL 107'-0"		
17	blind	T-101-N9	3/4"-P150-I-6032	Fit a blind between 3/4"-P150-I-6032/2 and the nozzle flange T-101-N9	E 18'-7" N 26'-6" EL 107'-0"		
18	valve	3/4"-P150-I-6033/2	3/4"-P150-I-6033	Close and lock VALVE, BALL, FLOATING 3/4" P150 (3/4"-P150-I-6033/2) on T-101-N10	E 18'-1/4" N 26'-6" EL 117'-0"		
19	blind	T-101-N10	3/4"-P150-I-6033	Fit a blind between 3/4"-P150-I-6033/2 and the nozzle flange T-101-N10	E 18'-7" N 26'-6" EL 117'-0"		

Inspections

Id	Inspection	Criteria	Basis
WO-100-01-I1	Internal visual examination of the shell, floor, roof structure and nozzles	Findings recorded against the owner's integrity criteria	SW-NDE-4
WO-100-01-I2	Ultrasonic thickness at the 15 TMLs listed	Each reading above the owner's minimum wall	SW-WO-6
WO-100-01-I3	Floor scan for under-floor corrosion over the full floor	Owner's floor acceptance criteria	SW-WO-6

Thickness monitoring locations (SW-WO-6)

TML	Item	Location	Position	Nominal wall	R1	R2	R3	R4
WO-100-01-TML01	T-101	Nozzle neck T-101-N1	E 31'-7-13/16" N 26'-6" EL 101'-6"		top: ____	side: ____	bottom: ____	side: ____
WO-100-01-TML02	T-101	Nozzle neck T-101-N2	E 25'-0" N 22'-6" EL 123'-3-1/16"		top: ____	side: ____	bottom: ____	side: ____
WO-100-01-TML03	T-101	Nozzle neck T-101-N3	E 22'-0" N 26'-6" EL 123'-4-5/16"		top: ____	side: ____	bottom: ____	side: ____
WO-100-01-TML04	T-101	Nozzle neck T-101-N4	E 25'-0" N 26'-6" EL 123'-3-11/16"		top: ____	side: ____	bottom: ____	side: ____
WO-100-01-TML05	T-101	Nozzle neck T-101-N5	E 18'-6-7/8" N 26'-6" EL 101'-0"		top: ____	side: ____	bottom: ____	side: ____
WO-100-01-TML06	T-101	Nozzle neck T-101-N6	E 28'-0" N 26'-6" EL 123'-1-7/16"		top: ____	side: ____	bottom: ____	side: ____
WO-100-01-TML07	T-101	Nozzle neck T-101-N7	E 18'-6-1/8" N 26'-6" EL 102'-6"		top: ____	side: ____	bottom: ____	side: ____
WO-100-01-TML08	T-101	Nozzle neck T-101-N8	E 18'-6-1/8" N 26'-6" EL 104'-6"		top: ____	side: ____	bottom: ____	side: ____
WO-100-01-TML09	T-101	Nozzle neck T-101-N9	E 18'-7" N 26'-6" EL 107'-0"		top: ____	side: ____	bottom: ____	side: ____
WO-100-01-TML10	T-101	Nozzle neck T-101-N10	E 18'-7" N 26'-6" EL 117'-0"		top: ____	side: ____	bottom: ____	side: ____
WO-100-01-TML11	T-101	Floor centre	E 25'-0" N 26'-6" EL 100'-6"		centre: ____			
WO-100-01-TML12	T-101	Shell at 0.000" above the floor	E 25'-0" N 26'-6" EL 100'-6"		0°: ____	90°: ____	180°: ____	270°: ____
WO-100-01-TML13	T-101	Shell at 6'-0" above the floor	E 25'-0" N 26'-6" EL 106'-6"		0°: ____	90°: ____	180°: ____	270°: ____
WO-100-01-TML14	T-101	Shell at 12'-0" above the floor	E 25'-0" N 26'-6" EL 112'-6"		0°: ____	90°: ____	180°: ____	270°: ____
WO-100-01-TML15	T-101	Shell at 18'-0" above the floor	E 25'-0" N 26'-6" EL 118'-6"		0°: ____	90°: ____	180°: ____	270°: ____

Test — Service leak test

Description	Every opened joint checked at operating pressure on start-up
Basis	SW-TEST-2

Inspection and test plan (SW-ITP-1..3)

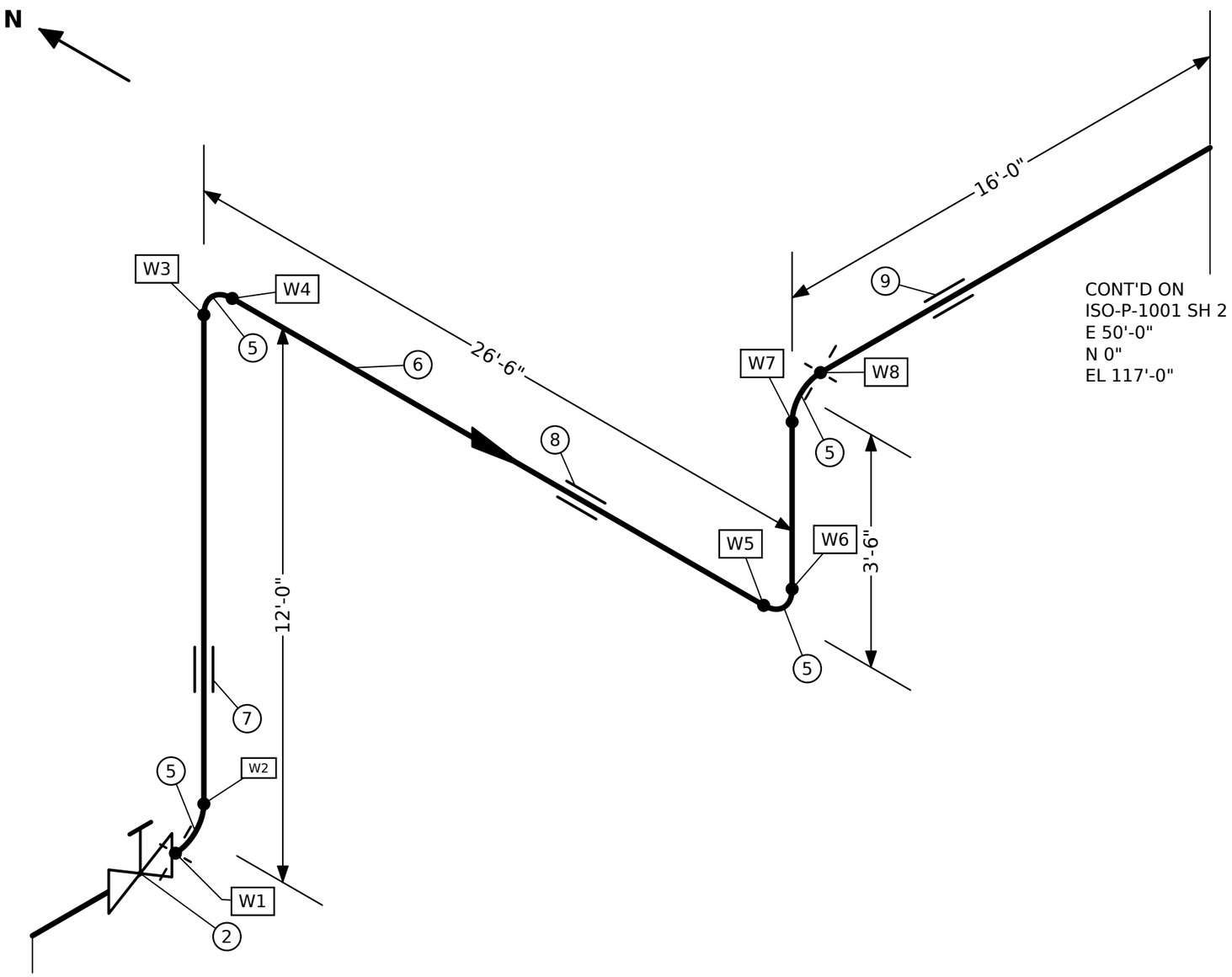
H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

#	Activity	Reference	Acceptance	Record	Contr.	Owner	3rd party	Status	Signed (contr. / owner / 3rd)	Date	Remark
1	Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained	SW-WO-3, isolation list	Every isolation in place and tagged; zero pressure at the vent	Signed isolation certificate	H	H	—				
2	Equipment vented, purged and gas-tested for entry; confined-space entry permit issued	Owner entry procedure	Atmosphere safe for entry per owner limits	Gas test record, entry permit	H	H	—				
3	Opened for entry; cleaned for inspection	Owner procedure	Surfaces clean enough to inspect	Daily report	S	S	—				
4	Internal visual examination	SW-NDE-4	Findings recorded	Inspection report with photographs	W	H	W				
5	Ultrasonic thickness at 15 TMLs	SW-WO-6	Above the owner's minimum wall	Thickness record per TML	S	R	R				
6	Floor scan	SW-WO-6	Owner's floor criteria	Floor scan report	S	R	R				
7	Owner's decision on findings (repair, monitor or accept)	Owner integrity procedure	Decision recorded	Signed disposition	R	H	R				
8	Closed with new gaskets on every opened joint; studs to preload	SW-WO-5	Preload as stated	Bolt-up record	W	W	—				
9	Service leak test of opened joints on start-up	SW-TEST-2	No leak	Leak test record	W	W	—				
10	Blinds removed, isolations returned to normal line-up per the isolation list	SW-WO-3	Every blind removed and accounted for	Isolation certificate closed	H	H	—				
11	Records complete; work order handed back to operations	SW-ITP-3	All ITP activities accepted or not applicable	Signed ITP and completion record	H	H	R				

Where the work is on the owner's drawings (SW-IMP-8)

Each affected item, replaced weld and isolation point, with the drawing and sheet that hold it and the weld label printed there. The sheets follow, copied unchanged.

Tag	Role	Drawing	Sheet	Weld label
6"-P150-P-1001/A2	isolation: valve	ISO-P-1001	1	
4"-P150-P-6038/2	isolation: valve	ISO-P-6038	1	
3"-P150-RV-4002/2	isolation: removal	ISO-RV-4002	1	
1"-P150-V-6030/2	isolation: valve	ISO-V-6030	1	
1"-P150-D-6031/2	isolation: valve	ISO-D-6031	1	
1"-P150-N-3015/B2	isolation: valve	ISO-N-3015	1	
2"-P300-LS-3007/B2	isolation: valve	ISO-LS-3007	1	
2"-P150-SC-3008/A2	isolation: valve	ISO-SC-3008	1	
3/4"-P150-I-6032/2	isolation: valve	ISO-I-6032	1	
3/4"-P150-I-6033/2	isolation: valve	ISO-I-6033	1	



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	2	6"	GASKET 6" P150 COMPRESSED SHEET
2	1	6"	VALVE, GATE, WEDGE, OS&Y 6" P150
3	2	6"	STUD SET 8 × 0.625" × 4" WITH 2 NUTS, FOR 6" P150
4	1	6"	FLANGE WELD-NECK 6" P150 RAISED FACE
5	4	6"	ELBOW 90° R = 5/3 OD 6" P150
6	52'-2"	6"	PIPE 6" P150 BEVELLED ENDS
7	1	6"	TRUNNION FOR 6" LINE (TRUNNION-6x15.0)
8	1	6"	HANGER FOR 6" LINE (HANGER-6x23.0)
9	1	6"	GUIDE FOR 6" LINE (GUIDE-6)

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-

COMPANY

SMARTWELD

TITLE

HYDROCARBON PROCESS

NOTES:

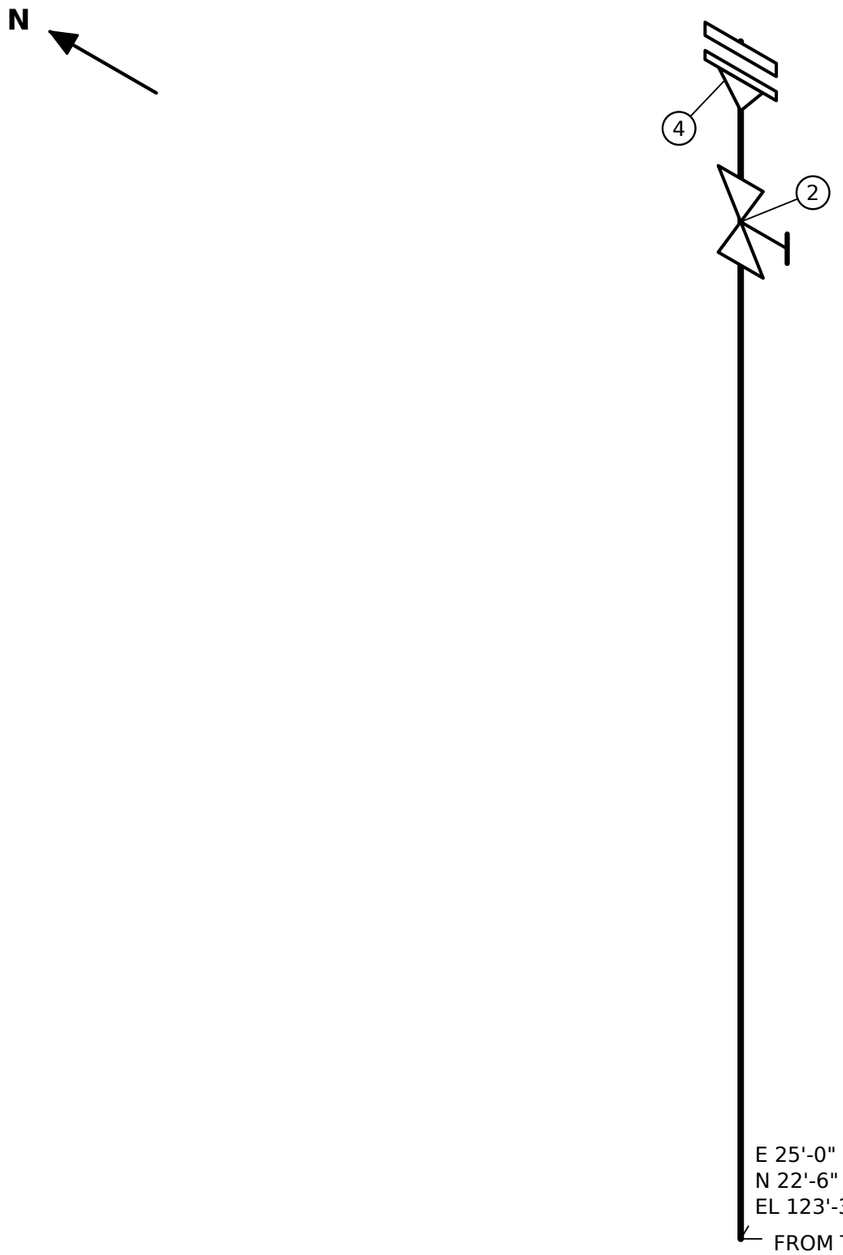
1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. HYDROTEST BEFORE PAINTING
4. REMOVE BURRS FROM ALL CUT ENDS

TEST PRESS.	INSULATION	PAINT
-	NONE	-
PWHT	NDE	FLUID
-	-	P

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
6"-P150-P-1001	H150	150 PSIG	-

DRAWING NO.	SHEET	REV
ISO-P-1001	1 OF 5	0

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	2	4"	GASKET 4" P150 COMPRESSED SHEET
2	1	4"	VALVE, GATE, WEDGE, OS&Y 4" P150
3	2	4"	STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P150
4	1	4"	FLANGE BLIND 4" P150 RAISED FACE

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-

NOTES:

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. FLANGE BOLT HOLES STRADDLE CENTRELINES
4. CHECK ORIENTATION BEFORE WELDING

TEST PRESS.	INSULATION	PAINT
-	NONE	-
PWHT	NDE	FLUID
-	-	P

COMPANY

SMARTWELD

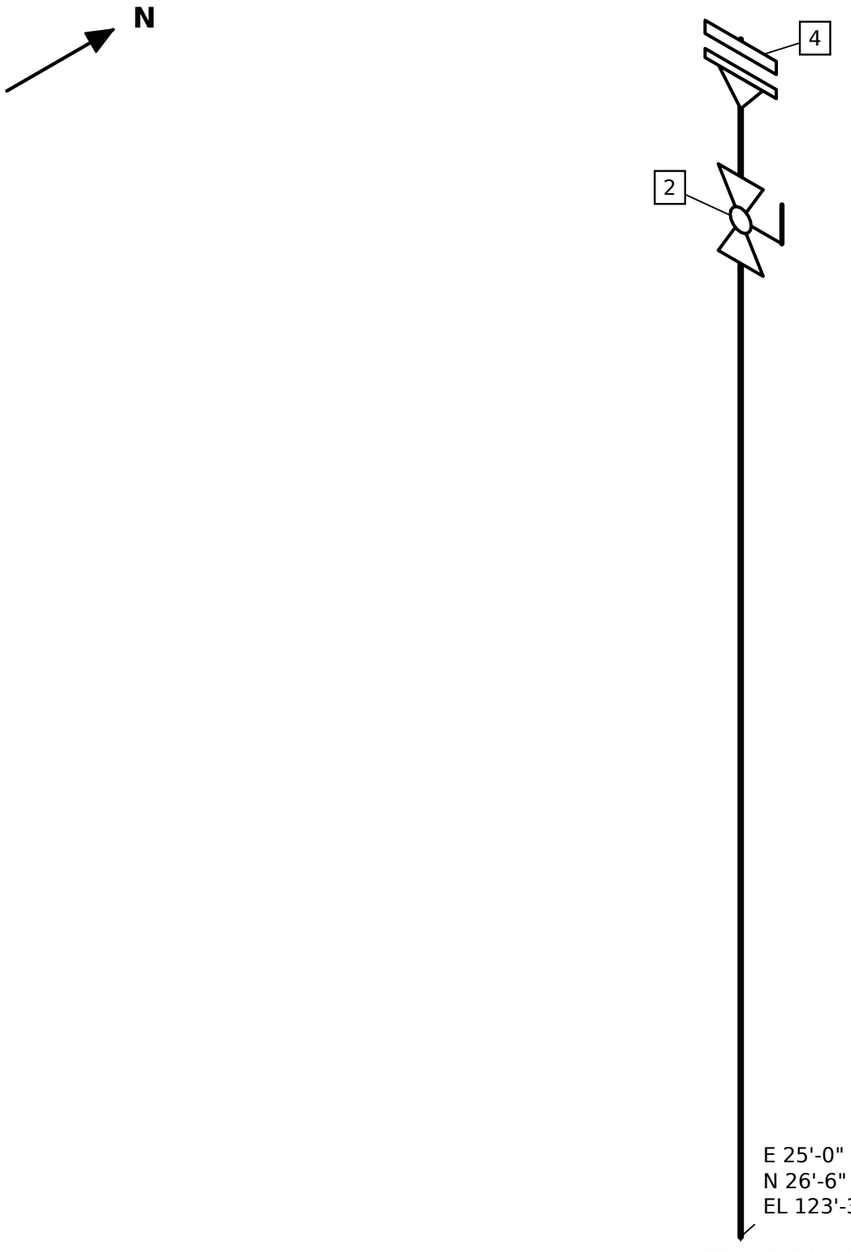
TITLE

HYDROCARBON PROCESS

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
4"-P150-P-6038	H150	150 PSIG	-

DRAWING NO.	SHEET	REV
ISO-P-6038	1 OF 1	0

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	2	1"	GASKET 1" P150 COMPRESSED SHEET
2	1	1"	VALVE, BALL, FLOATING 1" P150
3	2	1"	STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150
4	1	1"	FLANGE BLIND 1" P150 RAISED FACE

NOTES:

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. PAINT PER PROJECT SPECIFICATION

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-

COMPANY

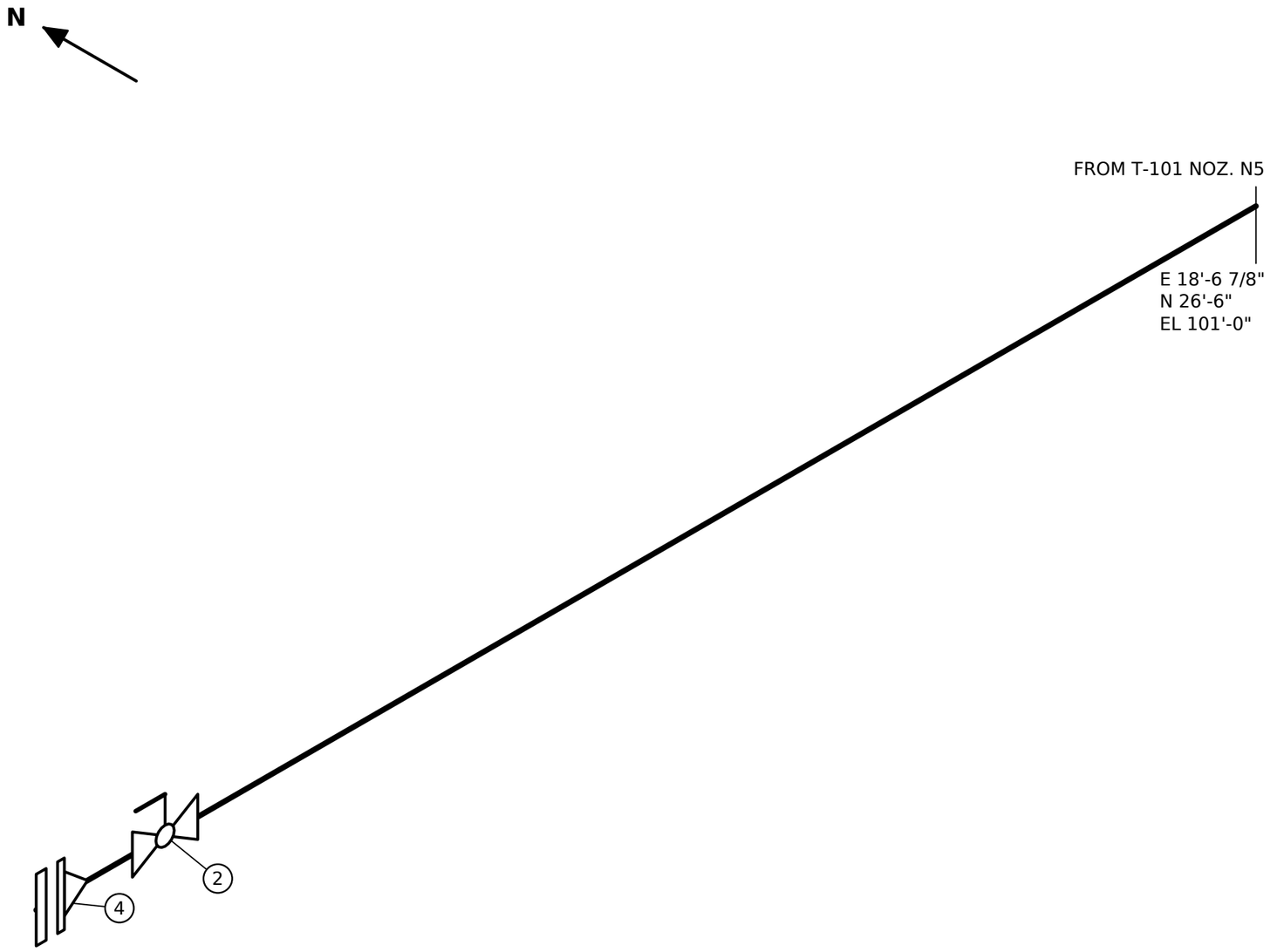
SMARTWELD

TITLE

HYDROCARBON PROCESS

TEST PRESS. **-**	INSULATION **NONE**	PAINT **-**	
PWHT **-**	NDE **-**	FLUID **V**	

LINE NO. **1"-P150-V-6030**	PIPING CLASS **H150**	DESIGN PRESS. **150 PSIG**	DESIGN TEMP. **-**
DRAWING NO. **ISO-V-6030**		SHEET **1 OF 1**	REV **0**
DATE **2026-11-01**	DRAWN **-**	CHECKED **-**	SCALE **NTS**



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	2	1"	GASKET 1" P150 COMPRESSED SHEET
2	1	1"	VALVE, BALL, FLOATING 1" P150
3	2	1"	STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150
4	1	1"	FLANGE BLIND 1" P150 RAISED FACE

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-

NOTES:

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. REMOVE BURRS FROM ALL CUT ENDS

TEST PRESS.	INSULATION	PAINT
-	NONE	-
PWHT	NDE	FLUID
-	-	D

COMPANY

SMARTWELD

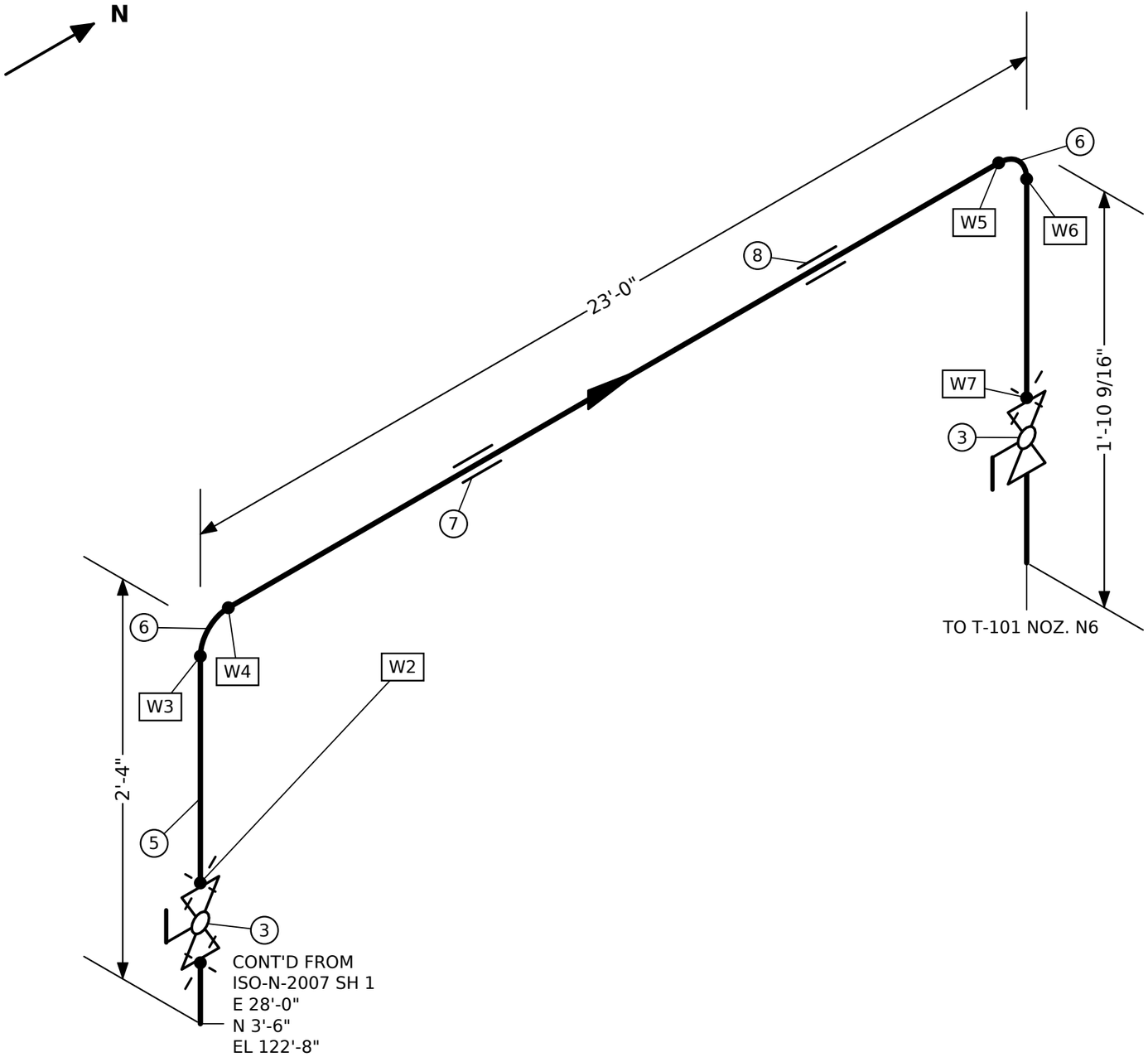
TITLE

HYDROCARBON PROCESS

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
1"-P150-D-6031	H150	150 PSIG	-

DRAWING NO.	SHEET	REV
ISO-D-6031	1 OF 1	0

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	3	1"	FLANGE WELD-NECK 1" P150 RAISED FACE
2	4	1"	GASKET 1" P150 COMPRESSED SHEET
3	2	1"	VALVE, BALL, FLOATING 1" P150
4	4	1"	STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150
5	23'-4 1/8"	1"	PIPE 1" P150 BEVELLED ENDS
6	2	1"	ELBOW 90° R = 5/3 OD 1" P150
7	1	1"	HANGER FOR 1" LINE (HANGER-1x88.0)
8	1	1"	HANGER FOR 1" LINE (HANGER-1x78.1)

NOTES:

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. HYDROTEST BEFORE PAINTING
4. FLANGE BOLT HOLES STRADDLE CENTRELINES

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-

COMPANY

SMARTWELD

TITLE

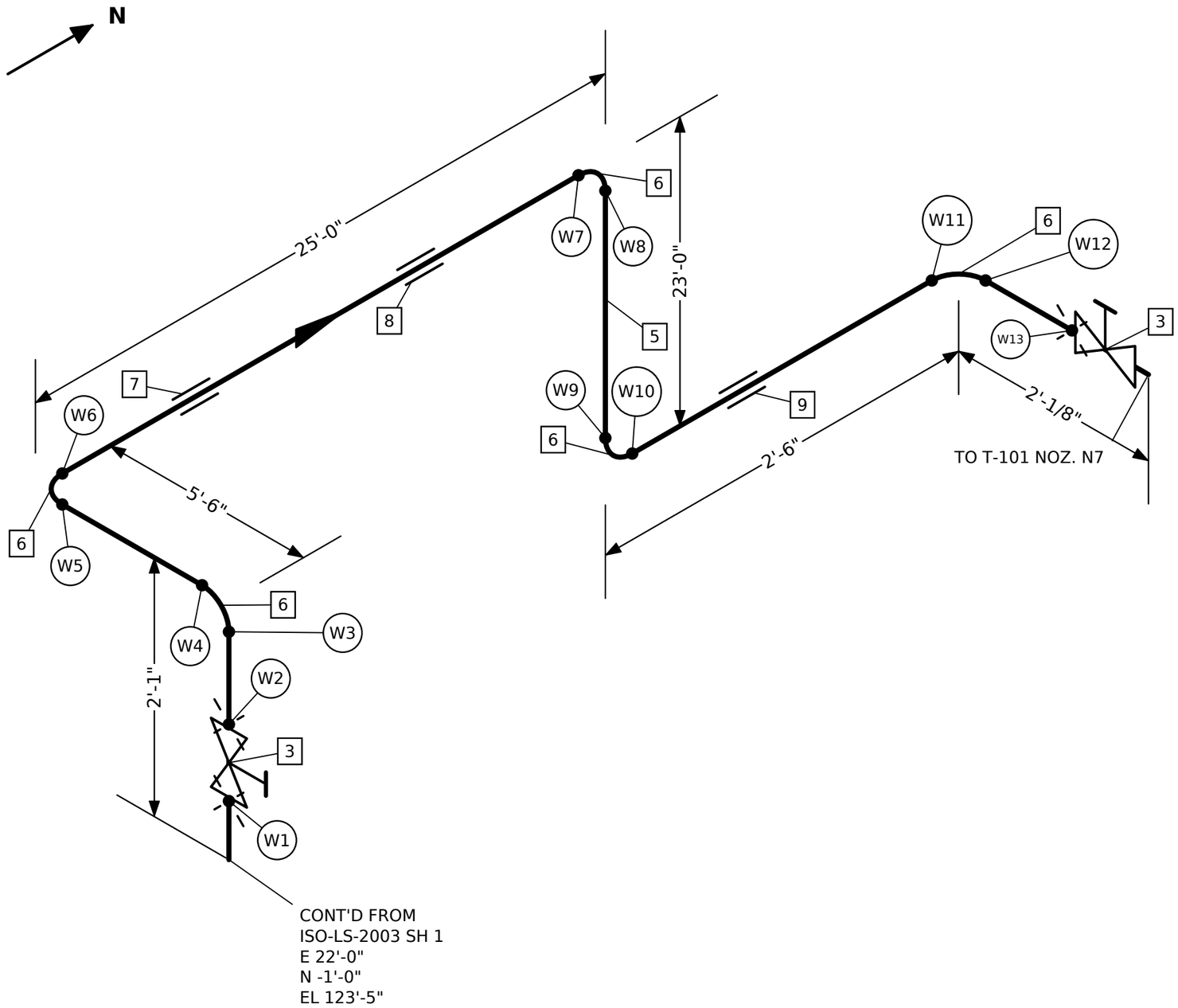
AIR AND GASES

TEST PRESS.	INSULATION	PAINT
-	NONE	-
PWHT	NDE	FLUID
-	-	N

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
1"-P150-N-3015	G150	150 PSIG	-

DRAWING NO.	SHEET	REV
ISO-N-3015	1 OF 1	0

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	3	2"	FLANGE WELD-NECK 2" P300 RAISED FACE
2	4	2"	GASKET 2" P300 COMPRESSED SHEET
3	2	2"	VALVE, GATE, WEDGE, OS&Y 2" P300
4	4	2"	STUD SET 4 × 0.625" × 4-1/4" WITH 2 NUTS, FOR 2" P300
5	54'-5 1/4"	2"	PIPE 2" P300 BEVELLED ENDS
6	5	2"	ELBOW 90° R = 5/3 OD 2" P300
7	1	2"	HANGER FOR 2" LINE (HANGER-2x87.0)
8	1	2"	HANGER FOR 2" LINE (HANGER-2x77.0)
9	1	2"	TRUNNION FOR 2" LINE (TRUNNION-2x29.0)

NOTES:
1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. HYDROTEST BEFORE PAINTING
4. CHECK ORIENTATION BEFORE WELDING

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-

COMPANY

SMARTWELD

TITLE

LOW-PRESSURE STEAM

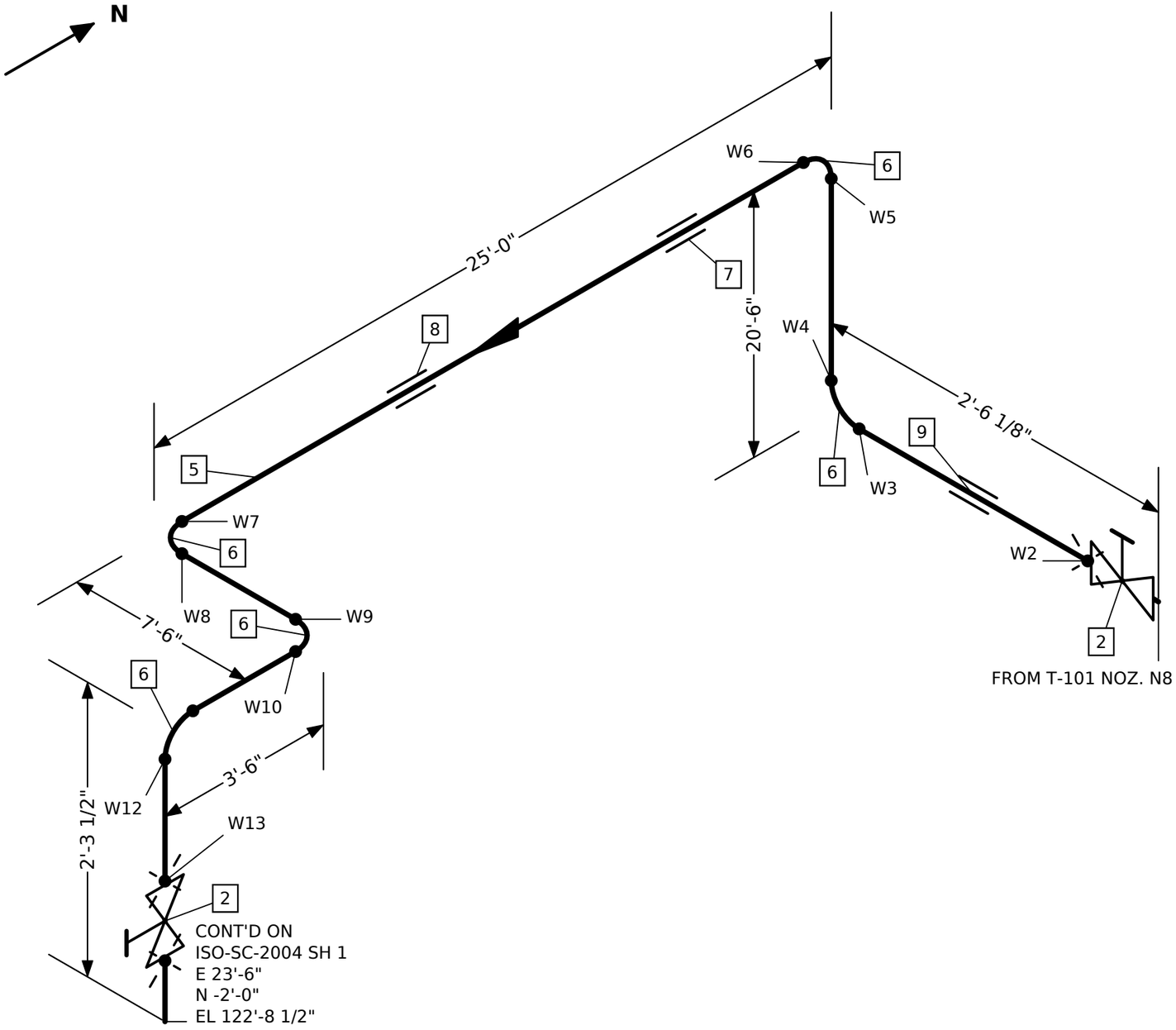
TEST PRESS.	INSULATION	PAINT
-	YES	-

PWHT	NDE	FLUID
-	-	LS

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
2"-P300-LS-3007	S300	300 PSIG	-

DRAWING NO.	SHEET	REV
ISO-LS-3007	1 OF 1	0

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	4	2"	GASKET 2" P150 COMPRESSED SHEET
2	2	2"	VALVE, GATE, WEDGE, OS&Y 2" P150
3	4	2"	STUD SET 4 × 0.625" × 4-1/4" WITH 2 NUTS, FOR 2" P150
4	3	2"	FLANGE WELD-NECK 2" P150 RAISED FACE
5	55'-7 3/4"	2"	PIPE 2" P150 BEVELLED ENDS
6	5	2"	ELBOW 90° R = 5/3 OD 2" P150
7	1	2"	HANGER FOR 2" LINE (HANGER-2x78.1)
8	1	2"	HANGER FOR 2" LINE (HANGER-2x88.0)
9	1	2"	TRUNNION FOR 2" LINE (TRUNNION-2x53.0)

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-

COMPANY

SMARTWELD

TITLE

STEAM CONDENSATE

NOTES:

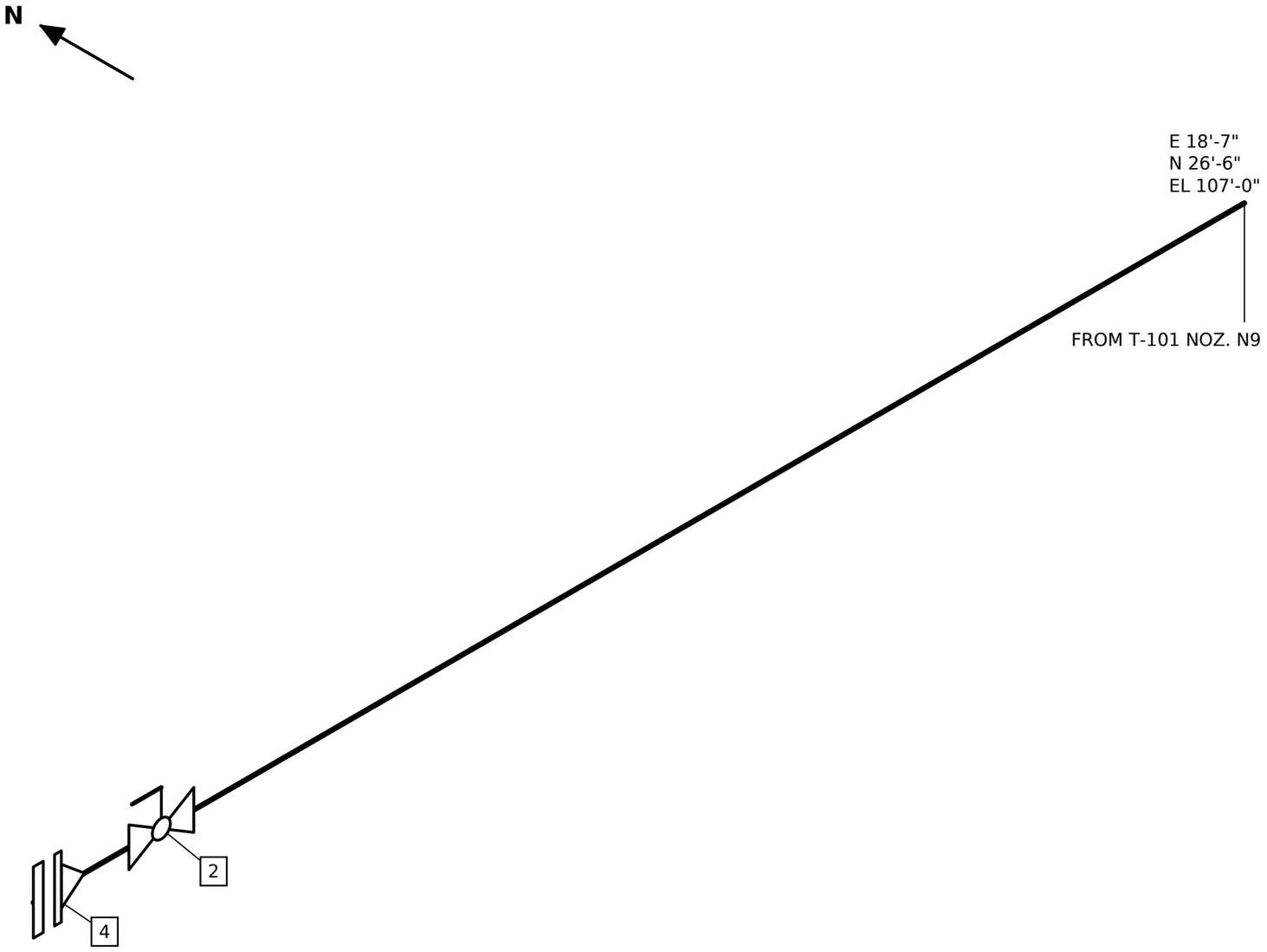
1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. HYDROTEST BEFORE PAINTING

TEST PRESS.	INSULATION	PAINT
-	YES	-
PWHT	NDE	FLUID
-	-	SC

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
2"-P150-SC-3008	C150	150 PSIG	-

DRAWING NO.	SHEET	REV
ISO-SC-3008	1 OF 1	0

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	2	3/4"	GASKET 3/4" P150 COMPRESSED SHEET
2	1	3/4"	VALVE, BALL, FLOATING 3/4" P150
3	2	3/4"	STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P150
4	1	3/4"	FLANGE BLIND 3/4" P150 RAISED FACE

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-

COMPANY

SMARTWELD

TITLE

HYDROCARBON PROCESS

NOTES:

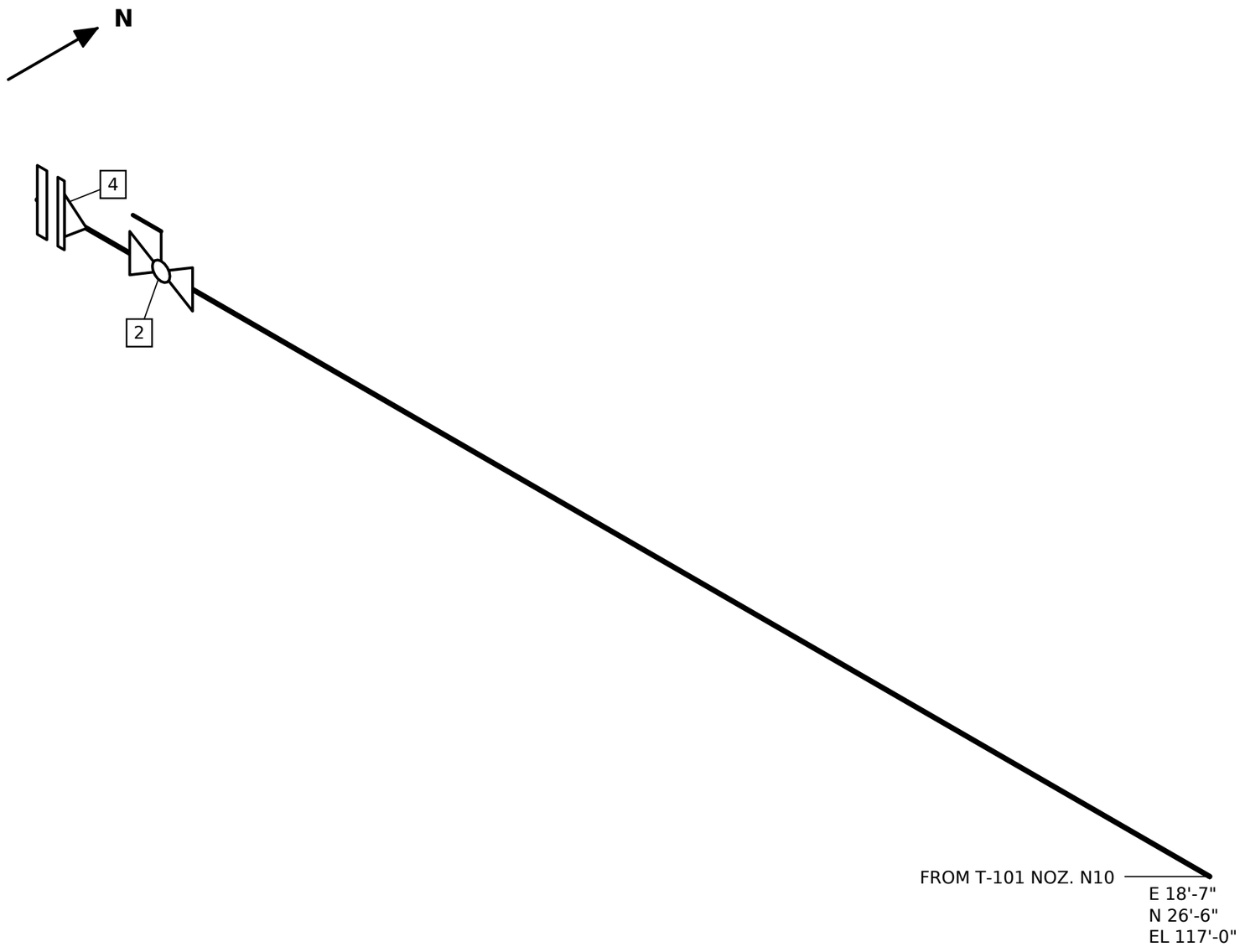
1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. HYDROTEST BEFORE PAINTING
4. REMOVE BURRS FROM ALL CUT ENDS

TEST PRESS.	INSULATION	PAINT
-	NONE	-
PWHT	NDE	FLUID
-	-	I

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
3/4"-P150-I-6032	H150	150 PSIG	-

DRAWING NO.	SHEET	REV
ISO-I-6032	1 OF 1	0

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	2	3/4"	GASKET 3/4" P150 COMPRESSED SHEET
2	1	3/4"	VALVE, BALL, FLOATING 3/4" P150
3	2	3/4"	STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P150
4	1	3/4"	FLANGE BLIND 3/4" P150 RAISED FACE

NOTES:

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. CHECK ORIENTATION BEFORE WELDING

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-

COMPANY

SMARTWELD

TITLE

HYDROCARBON PROCESS

TEST PRESS.	INSULATION	PAINT
-	NONE	-
PWHT	NDE	FLUID
-	-	I

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
3/4"-P150-I-6033	H150	150 PSIG	-

DRAWING NO.	SHEET	REV
ISO-I-6033	1 OF 1	0

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS

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